Physics of Semiconductor Devices

ECE210: PSD

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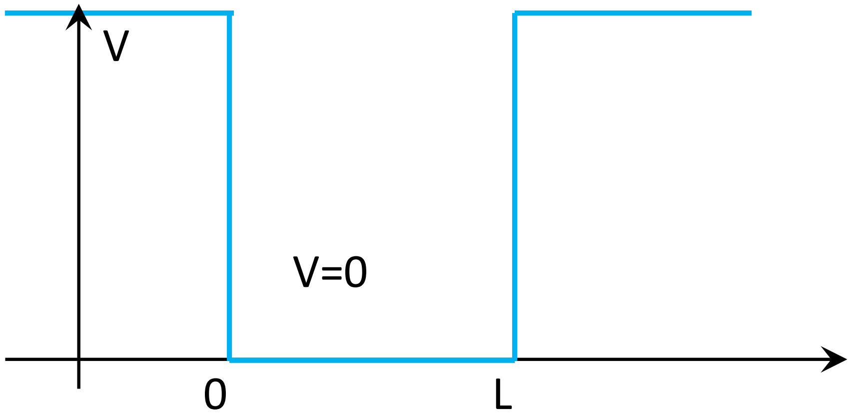
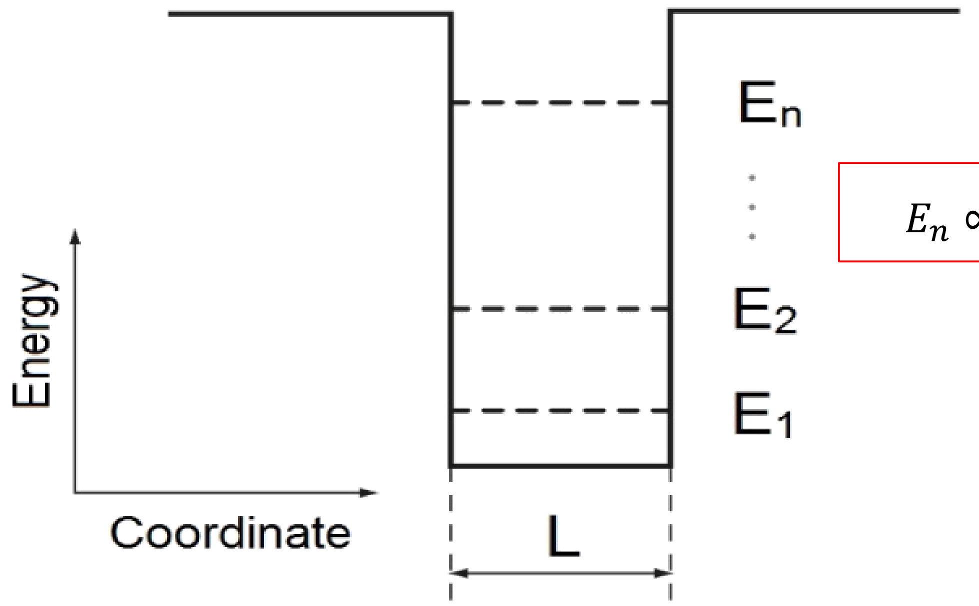
week 03

Particle in a Quantum well

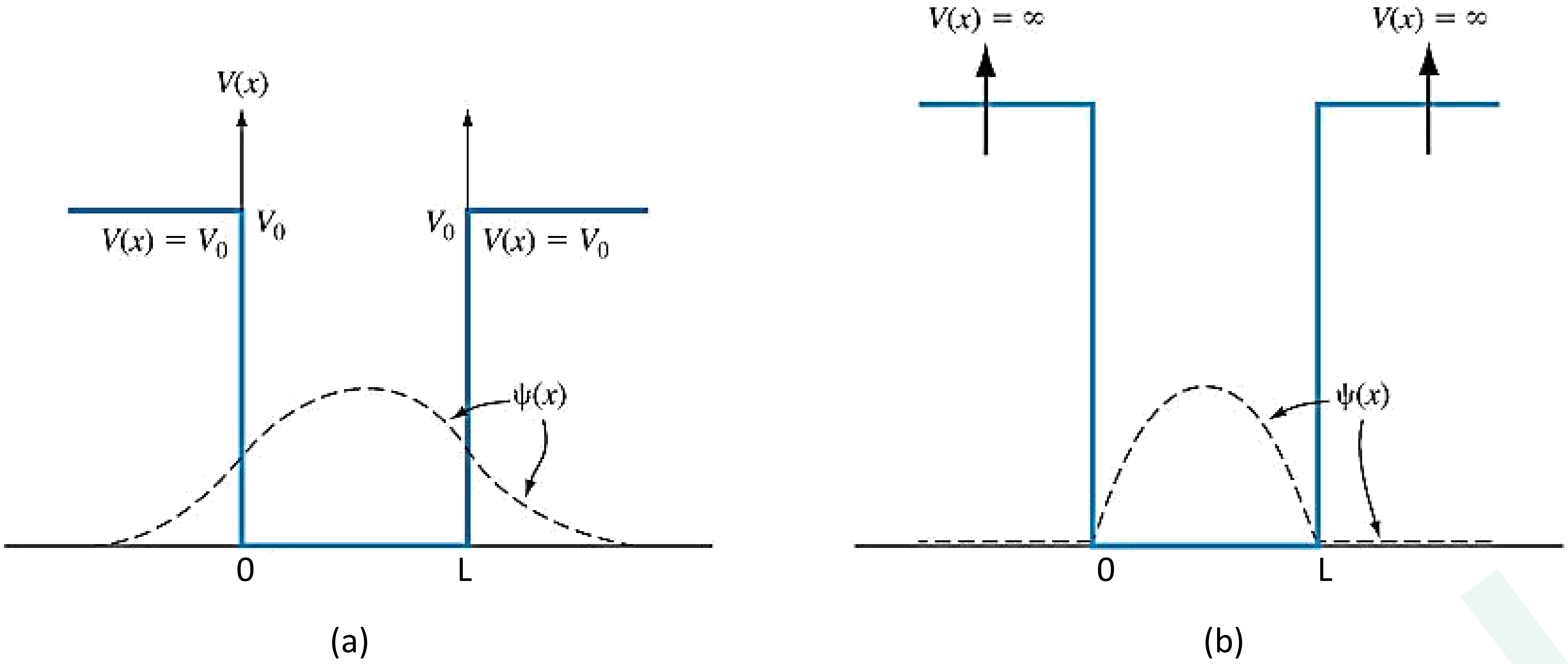




$$E_n \propto \left(\frac{n}{L}\right)^2$$

energy levels are discrete within a quantum well

contd.

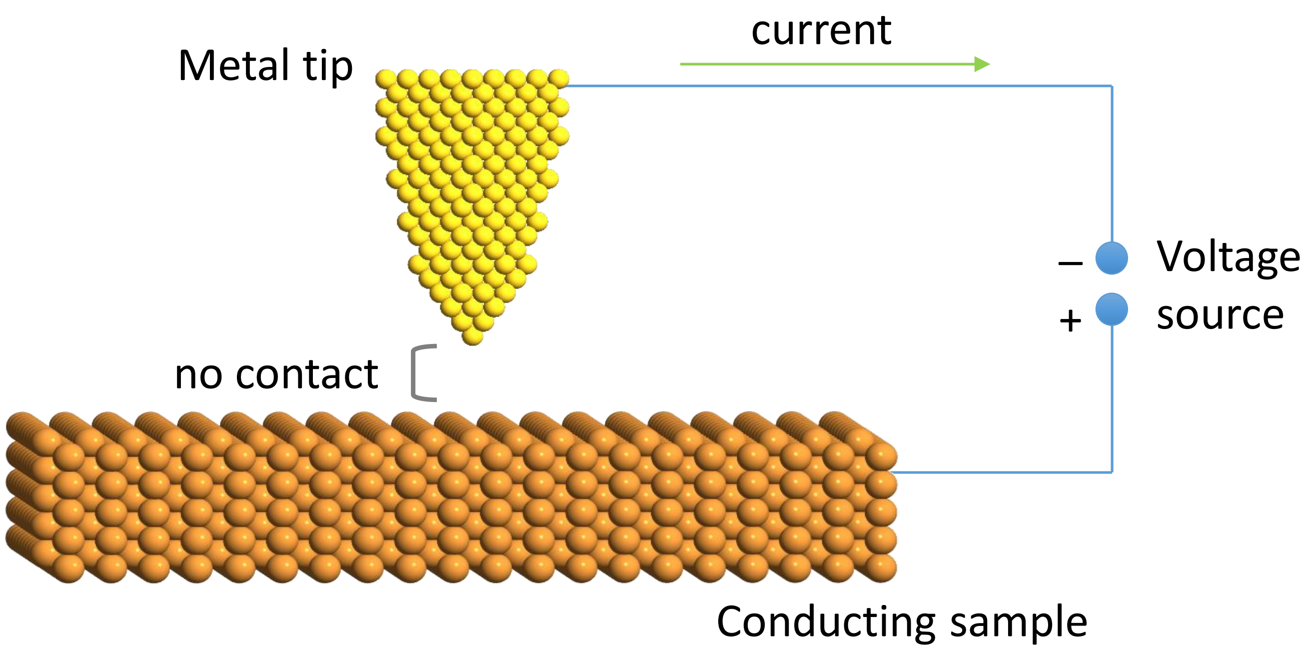


Potential function and corresponding wave function solutions for the case, when (a) the potential function is finite and (b) the potential function is infinite

Tunneling

Why tunneling:

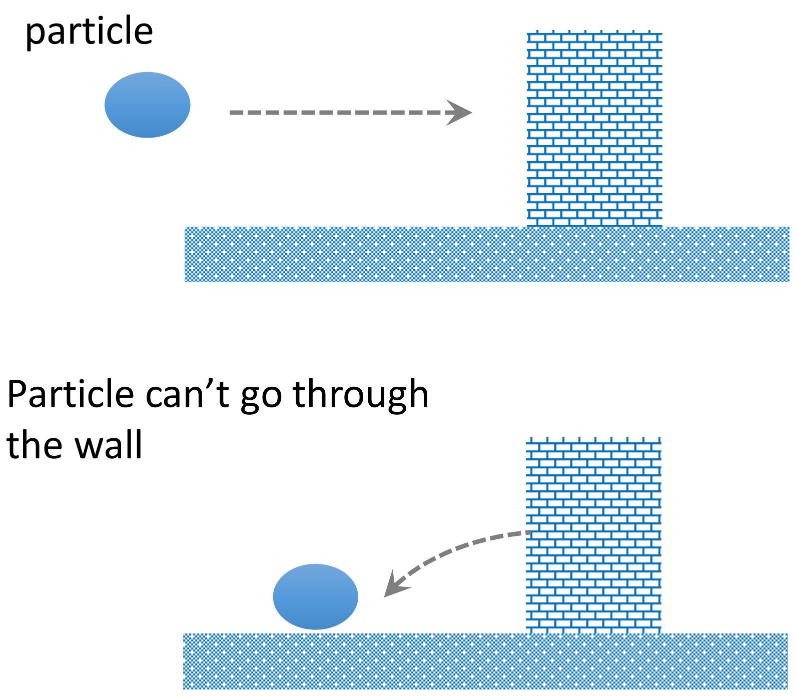
The concept of tunneling plays a significant role in the design and development of nanoelectronic devices. Whenever we connect nano objects like quantum dots to external leads, the concept of tunneling comes in to picture.



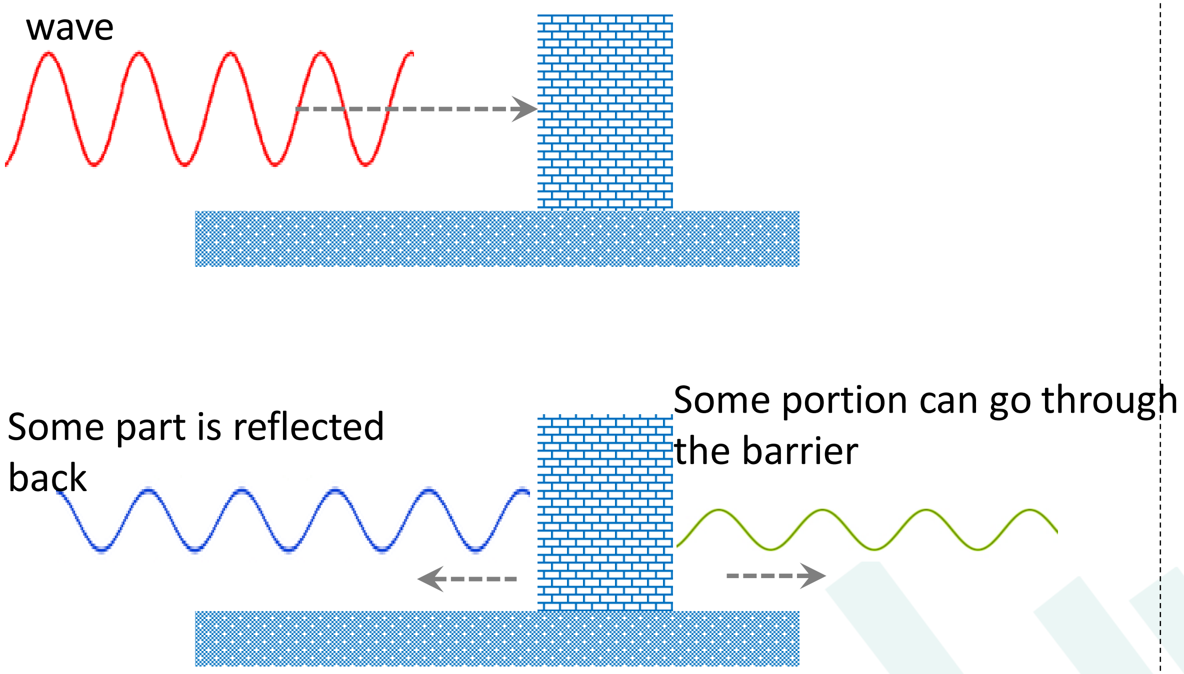
What is tunneling:

- **Tunneling is a quantum Mechanical phenomenon** with no analog in classical physics

Classical mechanics



Quantum mechanics



Schrodinger's equation (the building block for solving most of the elementary quantum problem)

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\Psi(x) = E\,\Psi(x)$$ where $\Psi \rightarrow$ the w.f.

Probability density: $\rho = \Psi^*\Psi$

Free particle solution: $\Psi_I \sim e^{ikx}$

contd.

The external potential term in the Schrodinger equation:

$$V(x) = \begin{cases} 0, x \leq 0 \\ V, x \in (0, L) \\ 0, x \geq L \end{cases}$$

Region I: the potential $V = 0$; hence Schrodinger's equation reduces to that of a free particle. Therefore, the its wave function

$$\Psi_I = \Psi_{I,i} + \Psi_{I,r} = Ae^{ikx} + Be^{-ikx} \qquad \text{where} \qquad k = \sqrt{\frac{2mE}{\hbar^2}}$$

Region II: finite potential barrier V

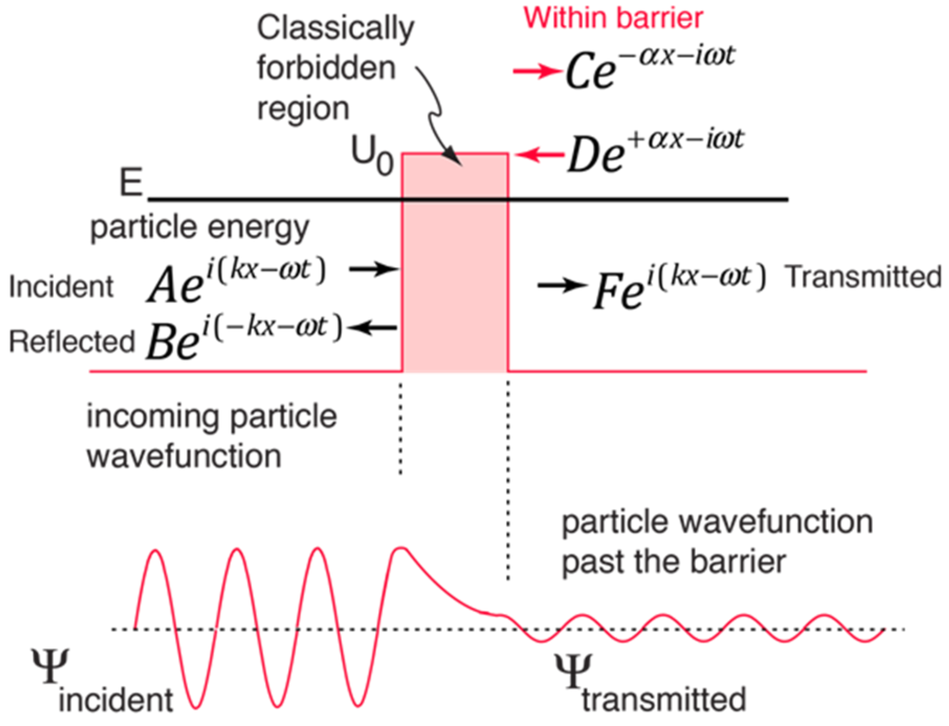
$$\frac{\hbar^2}{2m}\frac{\partial^2 \Psi_{II}}{\partial x^2} = (V - E)\Psi_{II}$$

Therefore, its wave function is

$$\Psi_{II} = Ce^{-k'x} + De^{k'x} \qquad \text{where} \qquad k' = \sqrt{\frac{2m(V-E)}{\hbar^2}}$$

Region III: the potential $V = 0$ and the transmitted particle is moving towards the right and has a positive momentum

$$\Psi_{III} = Fe^{ikx}$$



Transmission coefficient $T = \frac{(\Psi^*\Psi)_{transmitted}}{(\Psi^*\Psi)_{incident}} = \frac{|F|^2}{|A|^2}$

Tunneling probability:

$$T = \frac{|F|^2}{|A|^2} = \frac{1}{1 + \frac{V^2 sinh^2(k'L)}{4E(V-E)}}$$

contd.

Some interpretation from the variation of tunneling :

- **Probability increases** with increase in the particle energy.
- **'T' is greater for lower and thinner potential barriers**.
- The portion of tunneled particles as well as the **tunnel current can be gathered from the tunneling probability**.

Tunneling probability:

$$T = \frac{|F|^2}{|A|^2} = \frac{1}{1 + \frac{V^2 sinh^2(k'L)}{4E(V-E)}}$$

many practical tunneling problems cannot be approximated as tunneling through a rectangular potential barrier

(tunneling probability through a general potential barrier: potential "V" varies as a function of "x" in the previous formulation)

WKB approximation: (developed by Wentzel, Kramers and Brillouin)

$$T = \frac{|F|^2}{|A|^2} = e^{-2\gamma} \qquad \text{where} \qquad \gamma = \int_0^L \left| \sqrt{\frac{2m(V(x)-E)}{\hbar^2}} \right| dx$$